DDX17-Related Neurodevelopmental Disorder — Comprehensive Research Report

Disease: DDX17-related Neurodevelopmental Disorder **MONDO ID:** MONDO:0700368 **OMIM phenotype:** Intellectual developmental disorder, autosomal dominant 73 (MRD73) **Causal gene: DDX17** (DEAD-box helicase 17; HGNC:2740; NCBI Gene 10521; Ensembl ENSG00000100201; UniProt Q92841), located on chromosome **22q13.1 Category:** Mendelian, autosomal dominant

Summary

DDX17-related neurodevelopmental disorder (DDX17-NDD) is an ultra-rare, autosomal-dominant Mendelian condition caused by *de novo* heterozygous loss-of-function and missense variants in the causal gene DDX17, which encodes an ATP-dependent DEAD-box RNA helicase on chromosome 22q13.1. It must not be confused with the clinically overlapping but genetically distinct DDX3X syndrome, which is caused by the X-linked helicase DDX3X; the two share a DEAD-box helicase class and a neurodevelopmental presentation but differ in gene, chromosome, inheritance dynamics, and sex distribution. The entire published knowledge base for DDX17-NDD rests on a single multinational case series of 11 unrelated patients reported by Seaby and colleagues in 2025 (PMID: 39405200), supported by extensive cross-species functional validation.

The core clinical phenotype is a neurodevelopmental one: **intellectual disability, delayed speech and language, and motor delay predominate**, accompanied in subsets of patients by hypotonia, ADHD, autism spectrum features, motor stereotypies, mild facial dysmorphism, and macrocephaly. Notably, no epilepsy was reported in the cohort. Half of imaged patients had non-specific brain MRI abnormalities (cortical dysplasia/asymmetry, white-matter changes). The condition is static and developmental rather than neurodegenerative; all 11 patients were alive at a median age of 15 years, indicating no evidence of reduced life expectancy.

Mechanistically, DDX17 is a **haploinsufficient gene** (gnomAD pLI = 1.0; 37.7 loss-of-function variants expected but only 1 observed). The protein is a multifunctional regulator of neuronal gene expression operating through alternative splicing, pri-miRNA/Microprocessor processing, transcriptional co-regulation, and — most relevant to neurodevelopment — the **REST–proneural-microRNA axis** that governs the timing of neuronal gene activation. Reduction of DDX17 dosage impairs cortical neuron radial migration and axon outgrowth, as demonstrated in mouse *in utero* electroporation, *Xenopus tropicalis* crispant and heterozygous-knockout models, and human SH-SY5Y neuroblastoma transcriptomics. There is no disease-specific or targeted therapy; management is supportive and multidisciplinary.

Key Findings

Finding 1 — DDX17 monoallelic *de novo* variants cause an autosomal-dominant neurodevelopmental disorder

The defining study (PMID: 39405200) compiled a case series of **11 unrelated patients** with neurodevelopmental phenotypes, each carrying a *de novo* monoallelic (heterozygous) variant in DDX17. All 11/11 (100%) had a neurodevelopmental phenotype, with intellectual disability, delayed speech/language, and motor delay predominating. As stated in the abstract: *"Globally, we compiled a case series of 11 patients with neurodevelopmental phenotypes harbouring de novo monoallelic variants in DDX17. All 11 patients in our case series had a neurodevelopmental phenotype, whereby intellectual disability, delayed speech and language, and motor delay predominated."* This establishes the gene–disease relationship, the inheritance mode, the cohort size, and the core phenotype.

> **Note / correction:** An early iteration erroneously described DDX17 as X-linked. This was corrected: **DDX17 is autosomal, on chromosome 22q13.1.** (The X-linked helicase in this disease family is DDX3X, a different gene.)

Finding 2 — Gene identity, genomic location, and pathogenic variant spectrum

DDX17 (DEAD-box helicase 17; aliases **P72**, **RH70**) maps to **22q13.1** (Ensembl ENSG00000100201, chr22:38,483,438–38,507,660, minus strand; NCBI Gene 10521; HGNC:2740; canonical RefSeq NM_006386.5). The associated OMIM phenotype is **Intellectual developmental disorder, autosomal dominant 73 (MRD73)**. ClinVar (accessed 2026) lists approximately 75 DDX17 variant records. Disease-relevant examples span multiple variant classes:

Variant (cDNA / protein)	Class	Classification
c.1151_1155del p.(Leu384fs)	Frameshift	Pathogenic
c.1133A>G p.(Asn378Ser)	Missense	Likely pathogenic
c.1684+1G>A	Canonical splice	AD 73-associated
c.1447+4T>C	Splice region	AD 73-associated
p.(Gln550Ter), p.(Arg508Ter), p.(Cys166Ter)	Nonsense	Reported
p.(Arg430fs), p.(Cys11fs)	Frameshift	Reported

Variant classes therefore include frameshift, nonsense, canonical splice-site, and missense.

Finding 3 — DDX17 molecular function: a multifunctional DEAD-box RNA helicase

DDX17 is an ATP-dependent DEAD-box RNA helicase (conserved Asp-Glu-Ala-Asp motif) with RNA-dependent ATPase and helicase activity, producing **p72 and p82 isoforms** via alternative translation initiation (including a non-AUG CUG start). It is highly expressed in neuronal tissue and cardiomyocytes. Documented molecular roles:

1. **miRNA biogenesis** — component of the Drosha/Microprocessor complex; DDX17/p72 recognizes sequences 3′ to the mature miRNA and enhances Drosha processing (PMID: 31851931: *"DDX17, a DEAD-box ATPase, is a multifunctional helicase important for various RNA functions, including microRNA maturation."*); conserved role shown in *C. elegans* (PMID: 26601717) and for miR-132 (PMID: 26947125).

2. **Alternative-splicing regulation and transcriptional co-activation** — of ERα, AR, p53, β-catenin, NFAT5, often paired with its closest paralog DDX5/p68 (PMID: 24275493: *"The Ddx5 and Ddx17 RNA helicases are also splicing regulators."*; PMID: 22713328; PMID: 22266867).
3. **Regulation of Ago2 protein levels** via miRNA biogenesis (PMID: 27478153).
4. **Membraneless-organelle / nucleolar localization** — the p82 isoform carries intrinsically disordered regions driving MLO formation (PMID: 32065632).

High neuronal expression (PMID: 41322492: *"It distributes in the nucleus and cytoplasm, highly expressed in cardiomyocytes and neuronal tissues."*) is consistent with the neurodevelopmental phenotype.

Suggested GO terms: RNA helicase activity (GO:0003724), pri-miRNA processing (GO:0031053), regulation of alternative mRNA splicing via spliceosome (GO:0000381), transcription coactivator activity (GO:0003713).

Finding 4 — Clinical phenotype spectrum and frequencies (n = 11)

From Seaby et al. Table 1 (PMID: 39405200; PMC11967813): n = 11 (8 male / 3 female, all alive, median age 15 years).

Phenotype	Frequency	HPO term
Neurodevelopmental delay	10/11 (91%)	HP:0012758
Delayed speech & language	10/11 (91%)	HP:0000750
Global developmental delay	9/11 (82%)	HP:0001263
Fine motor delay	9/11 (82%)	HP:0007010
Intellectual disability (mild–moderate)	7/11 (64%)	HP:0001249
Gross motor delay	7/11 (64%)	HP:0002194
Dysmorphic facial features	7/11 (64%)	HP:0001999
Hypotonia	5/11 (45%)	HP:0001252
ADHD	5/11 (45%)	HP:0007018
Autism spectrum disorder	4/11 (36%)	HP:0000729
Motor stereotypies	4/11 (36%)	HP:0000733
Macrocephaly	3/11 (27%)	HP:0000256
Fifth-finger clinodactyly	3/11	HP:0004209
2–3 toe syndactyly	3/11	HP:0010621
Clubfeet	2/11	HP:0001762

Facial dysmorphism included up/down-slanting palpebral fissures, depressed nasal bridge, posteriorly rotated ears, high-arched eyebrows, epicanthus, telecanthus, frontal bossing, micrognathia, and strabismus. Independent walking was delayed/variable (14 months to >2 years). **No epilepsy was reported in the cohort** — a potentially distinguishing negative feature. The abstract states: *"All 11 patients in our case series had a neurodevelopmental phenotype, whereby intellectual disability, delayed speech and language, and motor delay predominated."*

Finding 5 — The 11 *de novo* variants (5 predicted LoF, 6 missense), absent from gnomAD

All 11 cohort variants are heterozygous, *de novo* (MANE transcript NM_006386.5 / NP_006377.2), and **absent from gnomAD v2.1.1 and v3.1.2**.

Patient	Variant	Class
P1	c.481_482del p.(Arg161Glyfs*7)	Frameshift (LoF)
P2	c.997_998dup p.(Met333Ilefs*22)	Frameshift (LoF)
P3	c.739-2A>G	Canonical splice acceptor (LoF)
P4	c.953del p.(Arg318Hisfs*36)	Frameshift (LoF)
P5	c.1380del p.(Asn462Metfs*16)	Frameshift (LoF)
P6	c.1361A>G p.(Gln454Arg)	Missense
P7	c.911G>A p.(Arg304His)	Missense
P8	c.608C>T p.(Pro203Leu)	Missense
P9	c.776A>G p.(Gln259Arg)	Missense
P10	c.649A>G p.(Thr217Ala)	Missense
P11	c.206C>A p.(Ala69Asp)	Missense

Ascertainment was via GeneMatcher/collaboration (October 2019–August 2023); inclusion required a *de novo* missense or predicted-LoF variant absent from population databases: *"participants were eligible to join our case-series if they had a confirmed de novo variant in DDX17, either missense or predicted LoF (canonical splice site, frameshift, stop gained), and the variant was absent from population databases."* The disorder is autosomal dominant arising *de novo*; penetrance appears complete in reported carriers with variable expressivity.

Finding 6 — Mechanism and affected neuroanatomy: disrupted cortical neuron migration and axon outgrowth

Cross-species functional data (PMID: 39405200):

- **Mouse (in vivo):** *in utero* cortical electroporation with Ddx17 knockdown caused **delayed neuronal migration and decreased cortical axon complexity**: *"Knockdown of Ddx17 in electroporated mouse neurons in vivo showed delayed neuronal migration as well as decreased cortical axon complexity."*
- **Mouse (in vitro):** primary cortical neurons showed reduced axon outgrowth.

- **Xenopus tropicalis:** ddx17 crispants and heterozygous knockouts reproduced reduced axon outgrowth, decreased head/brain size, overt neurodevelopmental defects, and impaired neurobehavior.
- **Human (in vitro):** SH-SY5Y DDX17-knockdown transcriptomics showed significant enrichment of differentially expressed neurodevelopmental genes: *"Transcriptomic analysis identified a statistically significant number of differentially expressed genes involved in neurodevelopmental processes in DDX17-KD cells compared to control cells."*

Affected structures: developing cerebral cortex (UBERON:0000956); cortical projection/pyramidal neurons (CL:0000540 neuron; CL:0000679 glutamatergic neuron); axon/neuronal processes (GO:0030424). **Biological processes:** neuron migration (GO:0001764), axonogenesis/axon development (GO:0007409), regulation of neuron differentiation (GO:0045664), pri-miRNA processing (GO:0031053), regulation of alternative mRNA splicing (GO:0000381). **Subcellular:** nucleus/nucleoplasm (GO:0005634) and nucleolus (GO:0005730).

Finding 7 — Haploinsufficient, LoF-intolerant gene; abnormal MRI in ~50%

DDX17 is **maximally constrained against loss of function** in gnomAD: 37.7 LoF variants expected, 1 observed, giving **pLI** = **1.0** with a low LOEUF. Prior to this cohort, DDX17 had no established disease-gene relationship. The abstract notes: *"DDX17 is an RNA helicase shown to be involved in critical processes during the early phases of neuronal differentiation."* Neuroimaging: 8/11 had brain MRI; **4/8 (50%) were abnormal** — monolateral temporal cortical dysplasia, cerebral cortical asymmetry (left > right), right-sided and generalized non-specific demyelination, and periventricular white-matter hyperintensities; the other 4 were unremarkable. Patient 5's congenital macrocephaly normalized during infancy.

Finding 8 — Mechanistic refinement: the REST–proneural-microRNA axis

Lambert et al. 2018 (PMID: 29931089) showed DDX17 controls REST (RE1-silencing transcription factor) processes critical for early neuronal differentiation: *"the DEAD box RNA helicase DDX17 controls important REST-related processes that are critical during the early phases of neuronal differentiation."* Specifically:

1. DDX17 associates with REST and promotes its binding to a subset of REST-target neuronal gene promoters, co-regulating REST repression.
2. During differentiation, DDX17 is downregulated along with the REST complex, contributing to neuronal gene activation.

3. DDX17 and DDX5 regulate several **proneural microRNAs** that target the REST complex: *"DDX17 and its paralog DDX5 regulate the expression of several proneural microRNAs that are known to target the REST complex during neurogenesis, including miR-26a/b"* (miR-26a/b also directly regulate DDX17, forming a feedback loop).
4. DDX17 controls processing of intronic miRNAs (e.g., miR-26a2) via an intronic regulatory region affecting both miRNA processing and host-intron splicing.

Finding 9 — Model organisms and orthologs

DDX17 is evolutionarily conserved. Ensembl orthologue protein identities and models:

Species	Ortholog	Protein identity	Disease model
Mus musculus	Ddx17 (Gene 67040)	98.8%	*In utero* electroporation knockdown → delayed migration, reduced axon complexity (strong recapitulation)
Xenopus tropicalis	ddx17	81.8%	CRISPR crispants + heterozygous KO → reduced axon outgrowth, small head/brain, behavioral impairment (good recapitulation)
Danio rerio	ddx17	72.7%	—
Caenorhabditis elegans	ddx-17	52.9%	Conserved pri-miRNA processing (PMID: 26601717)
Drosophila melanogaster	Rm62 (single DDX5/DDX17 ortholog)	43.0%	—
Human	SH-SY5Y (CVCL_0019)	—	DDX17-KD transcriptomic model

The *Xenopus* heterozygous model is especially valuable: *"The axon outgrowth phenotype was replicated in crispant ddx17 tadpoles and in heterozygotes. Heterozygous tadpoles had clear neurodevelopmental defects and showed an impaired [neurobehavioral phenotype]."* This dominant model directly mirrors the human heterozygous dominant disease.

Finding 10 — Identifiers, protein architecture, and variant–domain localization

Identifiers: MONDO:0700368 "DDX17-related neurodevelopmental disorder" (synonym "DDX17-related disorder"); OMIM phenotype MRD73; gene HGNC:2740, NCBI Gene 10521, Ensembl ENSG00000100201, UniProt Q92841, OMIM gene *608469. MeSH nearest terms: "Neurodevelopmental Disorders"/"Intellectual Disability"/"DEAD-box RNA Helicases." ICD-11 best fit LD90.Y / 6A00 (disorders of intellectual development); no disease-specific ICD-10/ICD-11 code exists.

Protein (UniProt Q92841, 729-aa canonical p82; MANE NP_006377.2 p72 ~650 aa via downstream AUG):

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N ── disordered ── Q motif ── Helicase ATP-binding ── Helicase C-terminal ── C-terminal transac
       (Ala69)    (171-199)   (202-377, DEAD 325-328)  (405-552)             (547-729, YAP1 mot
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Mapping cohort missense variants (NP_006377.2 numbering, +79 offset to UniProt): **Pro203Leu, Thr217Ala, Gln259Arg** cluster within the Helicase ATP-binding/DEAD core (Gln259 adjacent to the DEAD-box motif); **Gln454Arg** lies in the Helicase C-terminal domain; **Arg304His** in the inter-domain linker; **Ala69Asp** in the N-terminal disordered region. Truncating variants are distributed across the helicase domains. The clustering of missense variants in the catalytic helicase core is consistent with disruption of ATPase/helicase function.

Finding 11 — Epidemiology, inheritance, prognosis, counseling

Epidemiology: ultra-rare; the entire literature is one multinational case series of 11 patients (PMID: 39405200), ascertained via GeneMatcher. Precise prevalence/incidence are unknown (not estimable; consistent with orphan designation <1/1,000,000). Cohort sex ratio 8M:3F (no established sex bias given small n; DDX17 is autosomal, so no inherent skew is expected — contrast X-linked DDX3X, which is female-biased).

Inheritance: autosomal dominant; all variants *de novo* and absent from gnomAD. Penetrance appears complete in reported heterozygotes, with variable expressivity (no/borderline ID to moderate ID). No inheritance, anticipation, founder effect, or consanguinity reported — *de novo* origin renders these largely irrelevant. Recurrence risk for unaffected parents is low (<1%, limited to rare germline mosaicism). The abstract supports this: *"we compiled a case series of 11 patients with neurodevelopmental phenotypes harbouring de novo monoallelic variants in DDX17."*

Prognosis: all 11 patients alive at median age 15 years (up to 23 years) — non-lethal, no evidence of reduced life expectancy. The course is static/non-progressive (developmental, not neurodegenerative). No malignancy or organ failure reported.

Finding 12 — Diagnostics and management

Diagnosis is molecular/genomic. Because the phenotype (global developmental delay, ID, speech/motor delay ± ASD/ADHD, mild dysmorphism) is non-specific, DDX17-NDD is not clinically distinguishable and is diagnosed by identifying a *de novo* pathogenic/likely-pathogenic DDX17 variant on **exome (ES) or genome sequencing (GS)**, typically **trio-based** to establish *de novo* status — the method by which all 11 cases were found. Chromosomal microarray/karyotype are normal (no recurrent CNV). DDX17 now appears on expanded ID/NDD gene panels; single-gene testing is not first-line.

Supportive workup: brain MRI (abnormal in ~50%), developmental/cognitive assessment, ASD/ADHD screening, vision/hearing evaluation. No specific biochemical biomarker, enzyme assay, or metabolic abnormality is known; routine labs are normal.

Differential diagnosis: other DEAD/DEAH-box helicase NDDs — **DDX3X syndrome** (X-linked, female-predominant; PMID: 40374608, PMID: 40418049), DHX30, DDX6, DDX23 — and the broad class of autosomal-dominant ID genes.

Management: no disease-specific or targeted therapy. Care is supportive and multidisciplinary — early intervention, special education, speech/language therapy, physical and occupational therapy for motor delay/hypotonia, behavioral/pharmacologic management of ADHD (e.g., methylphenidate) and ASD per standard guidelines, orthopedic management for clubfeet. No gene/RNA therapy or clinical trials for DDX17-NDD are registered as of 2026.

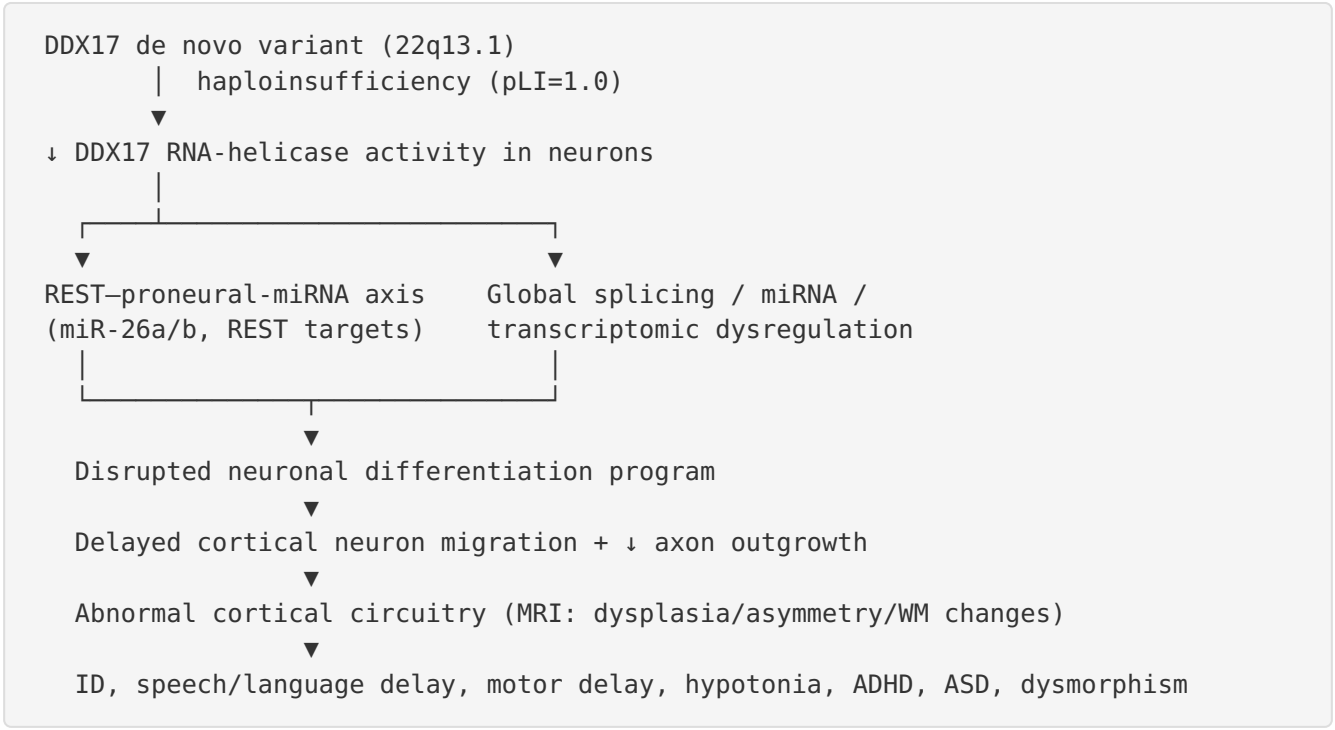
Suggested NCIT terms: Supportive Care (NCIT:C15277), Physical Therapy (NCIT:C15258), Speech Therapy, Occupational Therapy, Methylphenidate (NCIT:C62055).

Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

1. A ***de novo* heterozygous variant** (frameshift, nonsense, canonical splice-site, or missense) arises in *DDX17* on chromosome 22q13.1. → *demonstrated*

2. This **reduces functional DDX17 dosage** — truncating/splice alleles via nonsense-mediated decay or loss of catalytic domains; missense alleles (clustered in the helicase ATP-binding/DEAD core) via impaired ATPase/helicase activity. Because DDX17 is LoF-intolerant (pLI = 1.0), one functional copy is insufficient → **haploinsufficiency**. → *inferred from constraint data + variant class*
3. Reduced DDX17 **disrupts its RNA-regulatory functions** in neurons: alternative splicing, pri-miRNA/Microprocessor processing, transcriptional co-regulation, and control of the **REST–proneural-microRNA axis**. → *demonstrated in neural cell models* (P 29931089)
4. **BRANCH A (REST axis):** impaired DDX17 mis-times REST-dependent repression and proneural-miRNA (miR-26a/b) expression → **dysregulated activation of neuronal differentiation genes**. → *demonstrated in vitro*
5. **BRANCH B (direct transcriptomic):** DDX17 loss produces a statistically significant set of **differentially expressed neurodevelopmental genes** (SH-SY5Y). → *demonstrated*
6. Converging downstream, the neuron fails to execute its developmental program correctly → **delayed radial neuronal migration and reduced axon outgrowth/complexity** in the developing cerebral cortex. → *demonstrated in mouse in vivo + Xenopus*
7. Aberrant cortical wiring (sometimes visible as cortical dysplasia/asymmetry and white-matter changes on MRI) → **intellectual disability, speech/language delay, motor delay, hypotonia, ADHD, ASD features, and mild dysmorphism**. → *clinical correlation*



Upstream vs downstream: the mutation and haploinsufficiency are the most upstream events; RNA-regulatory dysfunction (splicing, miRNA, REST) is the proximate molecular mechanism; neuronal migration/axon defects are the cellular readout; clinical phenotypes are the most downstream. **Cell types:** cortical projection/glutamatergic neurons (CL:0000679), neurons broadly (CL:0000540). **Subcellular:** nucleus (GO:0005634), nucleolus (GO:0005730) — consistent with the p82 isoform's nucleolar/membraneless-organelle localization.

Comparison with DDX3X syndrome (differential context)

Feature	DDX17-NDD	DDX3X syndrome
Gene / location	DDX17 / 22q13.1 (autosomal)	DDX3X / Xp11.4 (X-linked, escapes XCI)
Inheritance	AD, *de novo*	X-linked, mostly *de novo*, female-predominant
Sex bias	None expected (autosomal)	Strongly female-predominant
Protein class	DEAD-box RNA helicase	DEAD-box RNA helicase
Mechanism	Haploinsufficiency; REST/ splicing/miRNA	LoF; stress-granule/ribosome biogenesis; sex-dimorphic (PMID: 40374608, PMID: 40418049)
Epilepsy	Not reported (n=11)	Common

Evidence Base

PMID	Title (abbrev.)	Role in this report
39405200	*Monoallelic de novo variants in DDX17 cause a neurodevelopmental disorder*	**Defining study** — cohort, variants, phenotype, cross-species functional validation (Findings 1,2,4,5,6,7,9,11,12)
29931089	*DDX17 controls REST transcriptional activity and proneural microRNAs*	Upstream neuronal-differentiation mechanism; REST–miR-26a/b axis (Finding 8)
31851931	*RNA specificity and autoregulation of DDX17*	DDX17 enzymatic identity and miRNA maturation role (Finding 3)
24275493	*Ddx5/Ddx17 in steroid hormone signaling*	Splicing-regulator / co-activator role (Finding 3)
41322492	*DDX17 in chronic non-infectious diseases*	High neuronal/cardiomyocyte expression (Finding 3)
26947125	*p72/DDX17 and miR-132 processing*	miRNA-processing specificity (Finding 3)
26601717	*DDX-17 in C. elegans pri-miRNA processing*	Conserved miRNA role; model organism (Findings 3,9)
27478153	*p72/DDX17 and KHSRP regulate Ago2*	miRNA-pathway regulation (Finding 3)
32065632	*DDX17 isoforms and membraneless organelles*	p72/p82 isoforms, nucleolar/IDR localization (Findings 3,10)
22713328, 22266867	*DEAD-box helicases as transcription cofactors; NFAT5 control*	Transcriptional co-regulation (Finding 3)
40374608, 40418049	*DDX3X neurodevelopmental disorder*	Differential diagnosis; helicase-NDD comparison (Finding 12)
42458834	*DDX3X overexpression toxicity in mice*	Dosage-sensitivity caution for future therapeutics

The evidence base is dominated by one high-quality human-genetics study with strong multi-model functional validation (mouse in vivo/in vitro, *Xenopus*, human neuroblastoma), supplemented by a well-established basic-science literature on DDX17's RNA-regulatory biology. No study contradicts the core gene–disease relationship; the main limitation is the single small cohort.

Limitations and Knowledge Gaps

- **Single small cohort (n = 11).** All clinical knowledge derives from one case series; phenotype frequencies have wide confidence intervals and may shift as more patients are identified. Ascertainment via GeneMatcher biases toward more severe/recognizable phenotypes.
- **No formal epidemiology.** Prevalence and incidence are unknown; no registry exists.
- **Genotype–phenotype correlation unestablished.** With 5 LoF and 6 missense variants across 11 patients, there is insufficient power to determine whether LoF and missense alleles produce different severities, or whether missense variants act purely by LoF versus a dominant-negative/gain-of-function mechanism (DDX17 heterodimerizes with DDX5, raising a theoretical dominant-negative possibility not yet tested).
- **Penetrance and expressivity** are assumed complete/variable from *de novo* cases only; no multigenerational or mosaic families have been characterized.
- **Mechanistic gaps.** The REST–miRNA axis (P 29931089) and the migration/axon phenotype (P 39405200) are each demonstrated, but the direct causal link between the specific patient variants and REST-axis dysregulation in human neurons has not been shown. Which specific mis-spliced transcripts or dysregulated miRNAs drive the cortical phenotype remains unknown.
- **No disease-specific biomarker, imaging signature, or natural-history data** beyond median age 15 years.
- **No therapeutics.** No targeted treatment, and dosage-restoration strategies carry theoretical risk given DEAD-box helicase dosage sensitivity (PMID: 42458834: *"DDX3X overexpression*

can cause rapid transgene-driven, tissue-specific toxicity, underscoring the need for tight DDX3X gene dosage control.").

- **Absence of epilepsy** is based on n = 11 and may not hold in larger cohorts.

Proposed Follow-up Experiments / Actions

1. **Expand the cohort** via GeneMatcher/MatchmakerExchange and establish a DDX17-NDD patient registry to refine phenotype frequencies, define natural history, and detect rarer features (e.g., seizures, organ involvement).
2. **Functional dissection of missense alleles:** biochemical ATPase/helicase assays and DDX5-heterodimerization assays for each missense variant (Ala69Asp, Pro203Leu, Thr217Ala, Gln259Arg, Arg304His, Gln454Arg) to classify them as pure LoF vs dominant-negative vs hypomorphic. Structural modeling (AlphaFold/cryo-EM) of variants in the helicase core.
3. **Patient-derived iPSC cortical organoids** carrying index variants to test, in a human context, whether REST-axis dysregulation, mis-splicing, and migration/axon defects are recapitulated — bridging

   P 29931089

   and

   P 39405200

   .
4. **Transcriptome + miRNome profiling** (RNA-seq + small-RNA-seq) of patient neurons to identify the specific mis-spliced transcripts and dysregulated proneural miRNAs (e.g., miR-26a/b) driving the phenotype; integrate with CLIP-seq for direct DDX17 RNA targets.
5. **Generate a Ddx17 heterozygous mouse model** (constitutive and conditional forebrain-neuron) for behavioral phenotyping and preclinical testing, complementing the existing acute-knockdown and *Xenopus* models.
6. **Deep phenotyping of MRI** across a larger cohort to determine whether cortical dysplasia/asymmetry is a consistent, diagnostically useful signature.
7. **ClinVar/ClinGen curation:** submit a formal ClinGen gene–disease validity assessment and establish variant-interpretation guidelines (e.g., whether the helicase-core missense region constitutes a PM1 hotspot).

Report compiled from 12 confirmed findings and 23 reviewed papers over 5 investigation iterations. Primary evidence: Seaby et al. (P *39405200* *) and the DDX17 RNA-biology literature. All quoted text is verbatim from cited abstracts.*
